

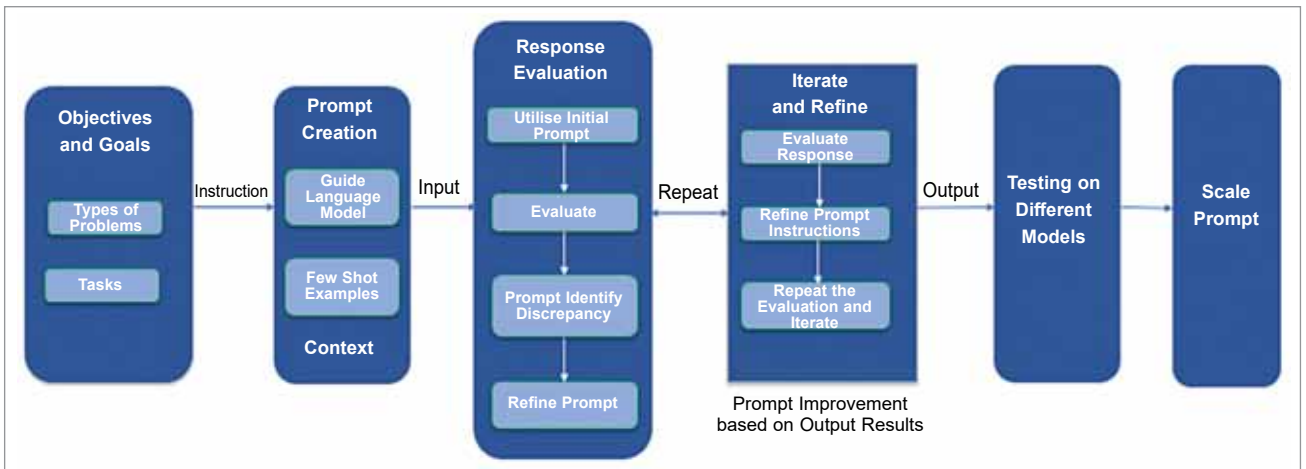


Fig. 1: Prompt engineering framework

Instruction. We must write clear instructions to the model for completing the task. An instruction, which can be simple or complex, does the following:

- Establishes the goal of the model
- Creates text that is being processed or transformed by the model

Context. Important details or context must be provided to get more relevant answers. The context does the following:

- Provides details that help the model to answer
- Assists the model with necessary information style and tone

Input data. Actual data that the model will be using is referred to as input data. Its requirements are:

- Express query as clearly as possible
- Query should be specific, in plain language, and no unnecessary fillers
- Include details in query to get more relevant answers
- Include examples of the desired behaviour of the model

Output indicator. This instructs the model about the desired output creation. The indicator does the following:

- Gives format guidance to model

- Supports content
- Continues to engineer the prompt until you achieve your desired results
- Assists model in generating more helpful solutions

Pillars of prompt engineering

The following are the pillars of prompt engineering techniques:

Examples. These act as a tool for fine-tuning LLM models. They help in getting the desired response.

Format. This describes user queries to receive the output results. It helps in generating productive outcomes from the LLM models.

Direction. This helps in narrowing responses for effective prompt output.

Parameters. These help in controlling the responses generated by LLM models.

Chaining. This connects multiple GenAI calls to complete the desired task.

Principles of prompt engineering

The main principles of prompt engineering to optimise responses are:

Clarity. Clearly communicate what content or information is most important. A good prompt

enables the LLMs to respond with more accuracy.

Structure the prompt. Start by defining its role, give context/ input data, and then provide the instruction.

Examples. Use specific examples to help the LLMs to narrow their focus and generate more accurate results.

Constraints. Use constraints to limit the scope of the model's output. This helps in receiving the desired response and avoids unwanted or unnecessary information.

Tasks. Break down complex tasks into a sequence of simpler and compact prompts. Make it simpler for LLMs to handle the data to comprehend each enquiry and generate meaningful answers.

Specific. Leave as little room for interpretation as possible. Restrict the operational space.

Prompt engineering framework

There are a few prompt engineering frameworks that perform prompt engineering tasks on AI models. These are Longchain, Semantic Kernel, Guidance, NeMo Guardrails, Fastrag, and LLamaIndex. Fig. 1 depicts the prompt engineering framework.